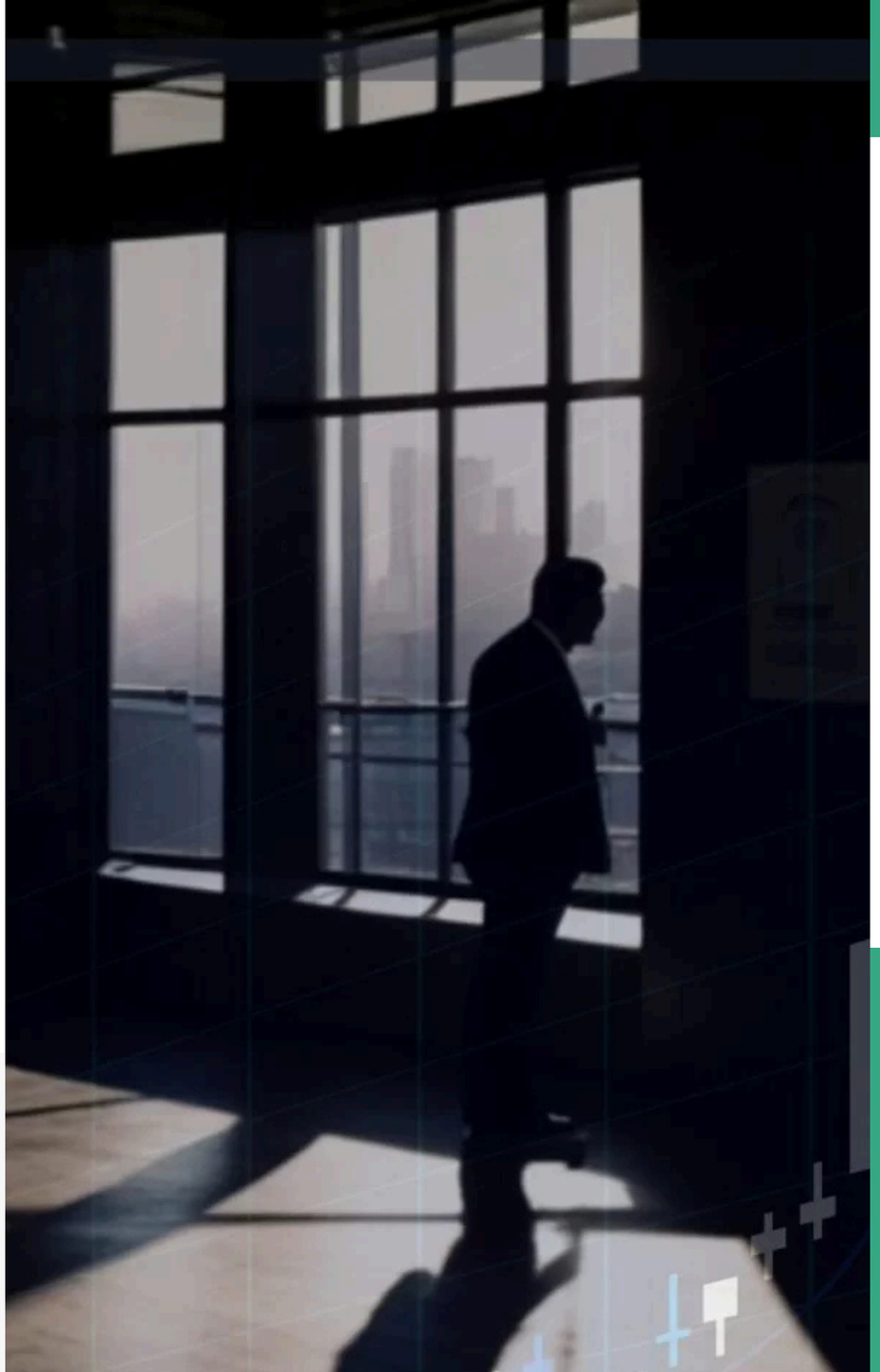
# SOLVING OPERATIONAL INEFICIENCIES IN COOX



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# INTRODUCTION

## About Coox

COOX is India's leading platform for booking home services like professional chefs, domestic cooks, bartenders, waiters, kitchen cleaners, live singers, appliances and crockery on rent, designer cakes, ingredients delivery and event decoration for occasions like house parties, weddings, corporate events, and more.

## The Challenge

- COOX regularly receives bookings from locations outside serviceable city areas.
- These bookings are ultimately cancelled and refunded, causing revenue loss.
- Customers experience poor service when orders are not fulfilled.
- No proactive mechanism exists to block these geographic zones.

## Objective

- Identify patterns in non-serviceable bookings
- Detect high-frequency geographic clusters
- Extract pin codes for problematic areas
- Build a geo-blocking recommendation

# KEY INSIGHTS

## Dataset

1. Given data frame contains following fields

| Booking ID | Payment Status | Address ID | Address Type | Area | City | State | Country | Lat | Long |
|---|---|---|---|---|---|---|---|---|---|
| 509103 | refunded_full | 152622 | Home | byalalu village chunchanaguppe post | Bengaluru | Karnataka | India | 12.897291 | 77.375508 |
| 700991 | refunded_full | 208569 | Farmhouse | Outside Hyderabad | Hyderabad | Telangana | India | 16.969071 | 78.518297 |
| 717282 | refunded_full | 216864 | Office | Aawasari Phata Manchar, Taluka Ambegaon, Mahar... | Pune | Maharashtra | India | 18.979685 | 73.947376 |

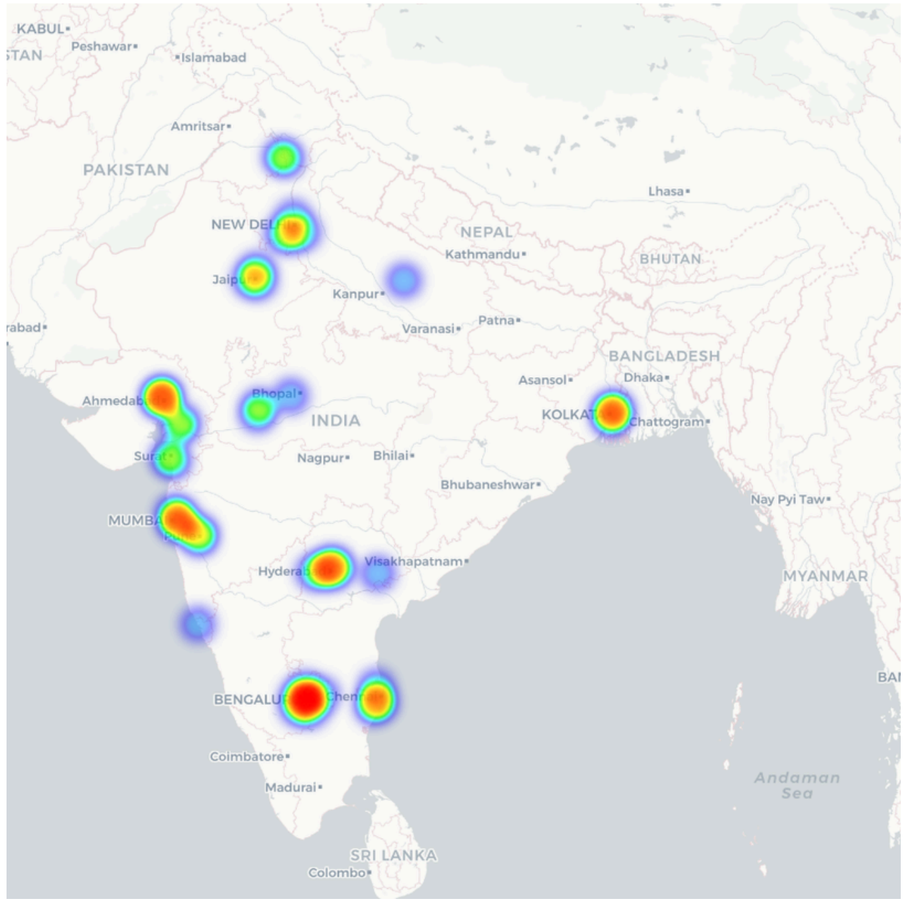
2. Shape of data is 557 rows × 10 columns
3. For all instances in data Payment Status is refunded_ful, so it do not contain data about severed locations

# EXPLORATORY DATA ANALYSIS

## Heat map of not full-filled bookings

Note: An interactive version of the same visualization has been uploaded with the report, titled "heatmap_unfulfilled_bookings."

In the heatmap, we can observe that unfulfilled orders are also originating from major cities or the outskirts of large urban areas, such as Delhi, Mumbai, Kolkata, Hyderabad, and Bangalore.



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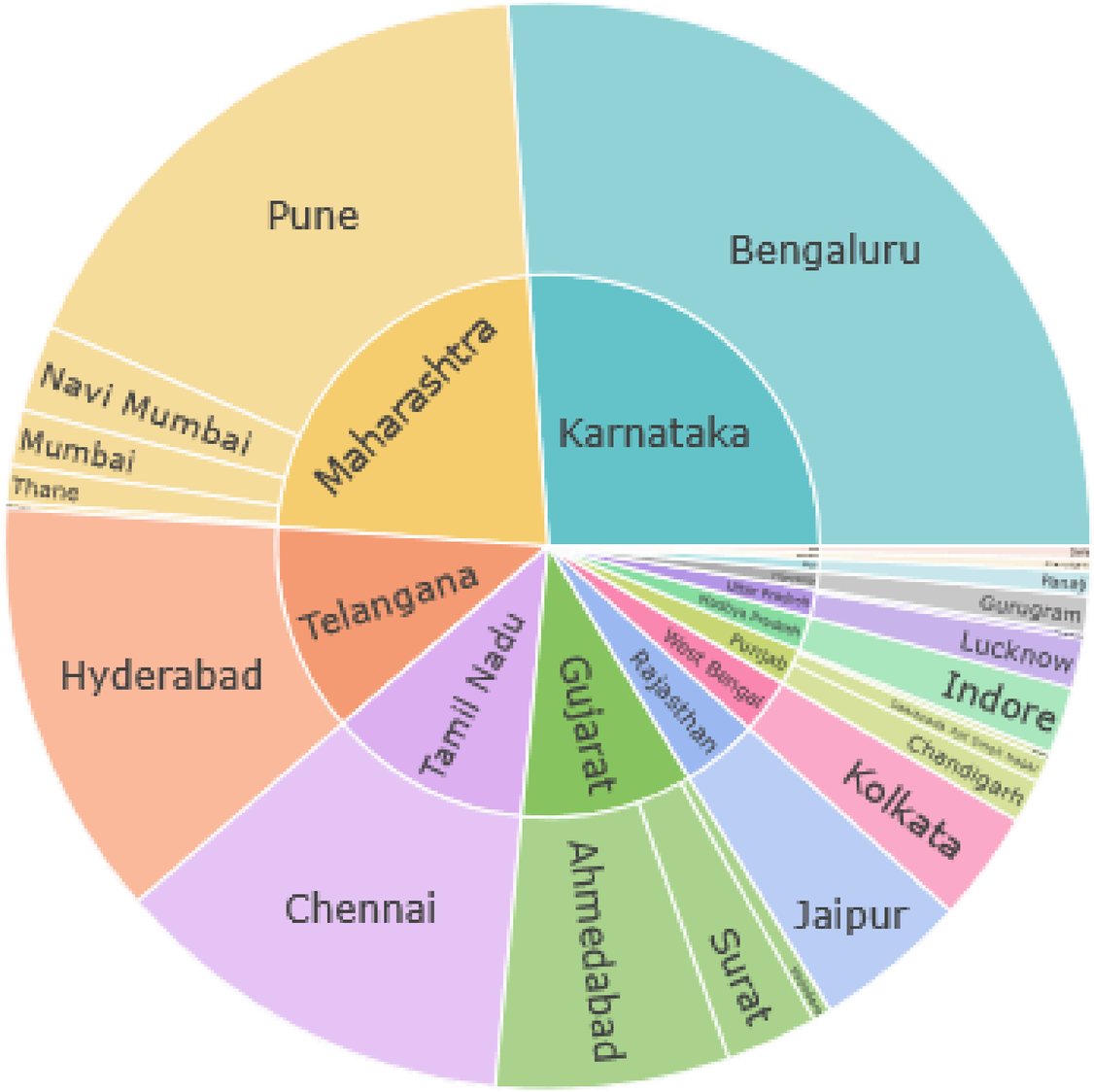
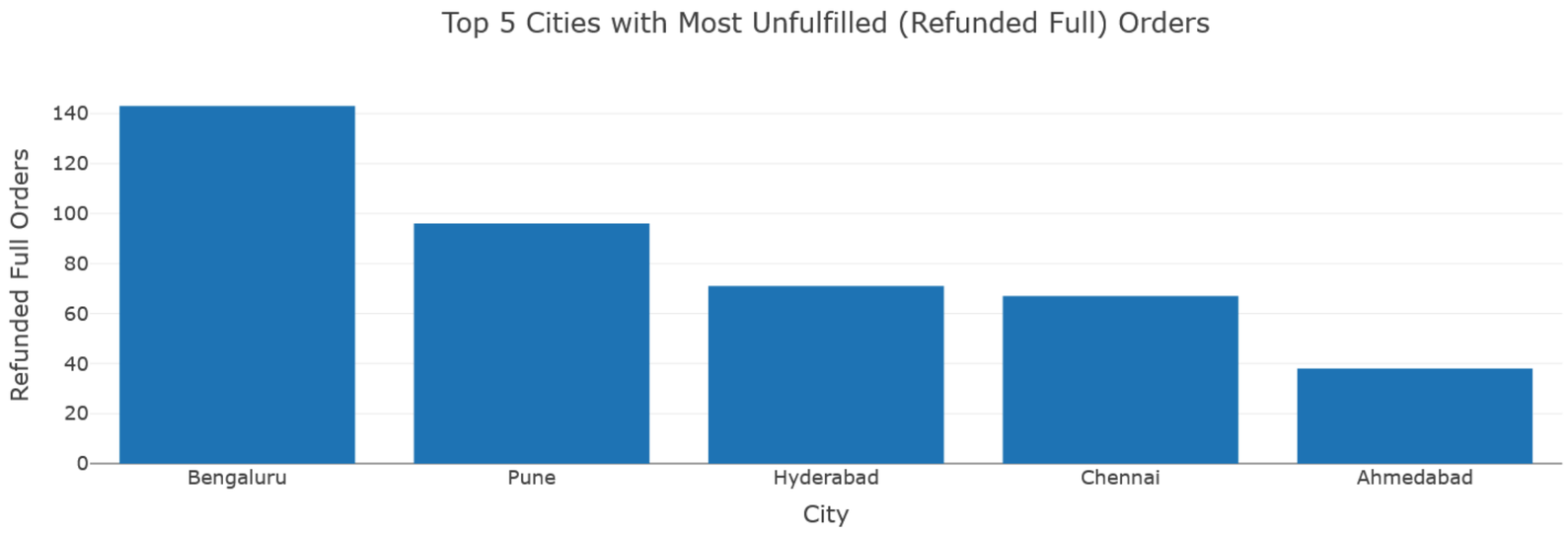
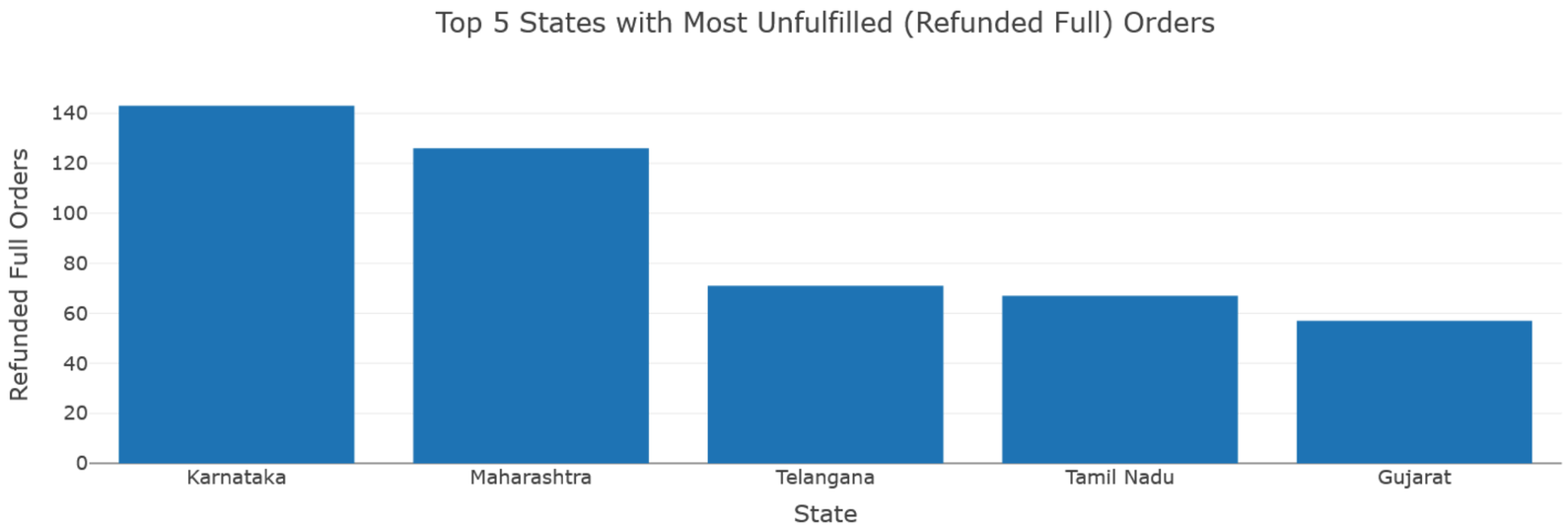



Geographic Breakdown of Outskirts Refunds

# EXPLORATORY DATA ANALYSIS



# EXPLORATORY DATA ANALYSIS

## Dealing with Null values

1. data contain 6 null value in Address type, 9 null value in area, 11 in Lat and 11 in Long.
2. we only ace about null value in Lat and Long, because we will be using DBSCAM algo for clustering.
3. good thing is all null values in Lat and Long have occurred together
4. so if we drop all row which have null value in Lat or Long column we only need to drop 11 rows out of 557 rows less than 2%. it will not cause any significant loss of info .

```
df.isna().sum()
```

| | 0 |
|---|---|
| Booking ID | 0 |
| Payment Status | 0 |
| Address ID | 0 |
| Address Type | 6 |
| Area | 9 |
| City | 0 |
| State | 0 |
| Country | 0 |
| Lat | 11 |
| Long | 11 |

# METHODOLOGY

- To identify non-servable clusters, we utilized Geospatial Clustering.
- Instead of traditional K-Means, the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm was used.

**Parameters:**

- epsilon (radius of circle used in DBSCAN) = 3 km
- min sample ( A point is considered a core point if it has a minimum number of other points within a given radius ε of itself) = 5

Distance Metric: The Haversine formula was applied to calculate the great-circle distance between two points on a sphere.

$$\text{distance} = 2r \arcsin\left(\sqrt{\sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)}\right)$$

# METHODOLOGY

## why we did not used K-mean clustering

1. You have to tell the number of clusters to be formed
2. Not good with arbitrary shaped clusters
3. Sensitive to outliers

**Reverse Geocoding**: Using the geopy library and Nominatim API, the centroids of the identified clusters were converted into actionable Pincodes

# RECOMMENDATIONS

We have identified 23 pin codes that are primarily situated on the outskirts of major cities such as Bangalore, Hyderabad, Mumbai, and others.from these locations high volumes are bookings are coming and company was not able to serve them

Note: An interactive version of the same visualization has been uploaded with the report, titled “Blocked_pincodes”

## Immediate Action

The immediate action that can be taken is to block those pin codes, or at the very least, restrict the top 10 to 15 pin codes with the highest number of unfulfilled bookings.

# SYSTEMATIC LONG-TERM PLAN

To avoid losing revenue and compromising customer satisfaction, it's essential to establish a comprehensive pipeline. This ensures the platform evolves alongside city growth and shifting demand patterns. The pipeline is structured as follows:

1. **Automated ML Detection:** Deploy a weekly pipeline to run DBSCAN clustering on new booking data, ensuring the system automatically detects emerging outskirts hotspots without manual oversight.
2. Transition from broad pincode-level blocking to precise coordinate-based geofences (circular or polygon boundaries) to avoid accidentally blocking serviceable customers.
3. Treat identified "refund hotspots" as a roadmap for service expansion, prioritizing partner recruitment in these high-demand areas to turn revenue leakage into new growth.
4. Establish a monthly review cycle to re-test and unblock areas as city infrastructure improves or operational limits expand, preventing the permanent exclusion of newly serviceable locations

# THANK YOU